

Quiz (Carolina Reaper)

- Q1.** What is the purpose of a stem-and-leaf plot?
(A) To display the mean of a dataset
(B) To show the distribution of a dataset
(C) To calculate the median of a dataset
(D) To find the mode of a dataset
(E) To graph a linear equation
- Q2.** Which of the following is a benefit of using a stem-and-leaf plot?
(A) It can only be used with small datasets
(B) It is difficult to read and interpret
(C) It can help identify patterns and trends in the data
(D) It is only used for categorical data
(E) It is not useful for comparing multiple datasets
- Q3.** What is the stem in a stem-and-leaf plot?
(A) The leaf of the plot
(B) The stem of the plot
(C) The first digit of a number
(D) The last digit of a number
(E) The title of the plot
- Q4.** What is the leaf in a stem-and-leaf plot?
(A) The stem of the plot
(B) The title of the plot
(C) The first digit of a number
(D) The remaining digits of a number
(E) The axis labels
- Q5.** Suppose we have the dataset {12, 13, 15, 17, 19, 20, 22, 25, 30}.
- What would be the stem-and-leaf plot for this dataset?
(A) Stem: 1, Leaves: 2, 3, 5, 7, 9
(B) Stem: 1, Leaves: 2, 3, 5, 7, 9, 0
(C) Stem: 1, Leaves: 2, 3, 5, 7, 9, 0, 2
(D) Stem: 2, Leaves: 0, 2, 5
(E) Stem: 2, Leaves: 0, 2, 5, 0
- Q6.** Which of the following datasets would have the most spread out stem-and-leaf plot?
(A) Dataset with values 1 – 10
(B) Dataset with values 10 – 20
(C) Dataset with values 100 – 200
(D) Dataset with values 1000 – 2000
(E) Dataset with values 10,000 – 20,000
- Q7.** How do you determine the number of stems in a stem-and-leaf plot?
(A) Count the number of leaves
(B) Count the number of digits in the largest number
(C) Determine the number of unique first digits
(D) Determine the number of unique last digits
(E) Count the number of data points
- Q8.** What can be inferred about a dataset if its stem-and-leaf plot is skewed to the right?
(A) The dataset has a large number of small values
(B) The dataset has a large number of large values
(C) The dataset is symmetric
(D) The dataset has a small range
(E) The dataset has a large median

Open Ended Questions (FRQ)

- Q9.** Construct a stem-and-leaf plot for the dataset {23, 25, 27, 30, 32, 35, 38, 40, 45, 50} and describe the distribution of the data.
- Q10.** Explain how to use a stem-and-leaf plot to compare the distribution of two datasets. Provide an example with two small datasets.

Chef's Tips

- Tip 1: Emphasize the importance of understanding the structure of a stem-and-leaf plot.
- Tip 2: Encourage students to practice constructing and interpreting stem-and-leaf plots with various datasets.
- Tip 3: Remind students to consider the shape and spread of the distribution when interpreting a stem-and-leaf plot.